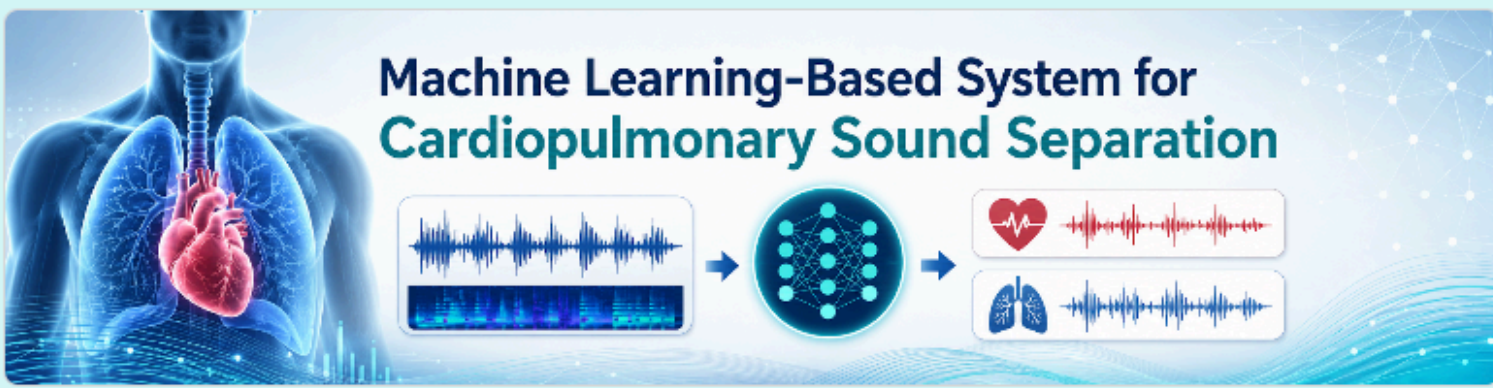




Machine Learning-Based System for Cardiopulmonary Sound Separation

User Requirements Questionnaire for Machine Learning-Based Cardiopulmonary Sound Separation System

Welcome and thank you for participating in this questionnaire.

This questionnaire is conducted as part of a Final Year Project at **Multimedia University (MMU)** involving the development of a machine learning-based system for cardiopulmonary sound separation. The proposed system aims to allow users to upload a mixed cardiopulmonary audio recording and obtain separated heart and lung sound outputs.

The purpose of this questionnaire is to understand user background, current challenges in working with biomedical or audio-processing tasks, and user preferences regarding system features, interface design, output presentation, and evaluation needs.

Your responses will help identify user requirements and support the design of the proposed prototype. There are no right or wrong answers. All responses will be used for academic purposes only and will remain confidential.

Estimated completion time: 3-5 minutes.

Section A - Respondent Background:

Tell us about your background and experience with audio, machine learning, or biomedical signals.

1. What is your age group? *

☐ Under 18

☐ 18-25

☐ 26-35

☐ 36 and above

2. What best describes you? *

- ☐ Software Engineering / Computer Science student
- ☐ AI / Machine Learning student
- ☐ Biomedical Engineering/ Health-related student
- ☐ Researcher
- ☐ Lecturer / Supervisor
- ☐ Other:

3. How familiar are you with audio processing or signal processing? *

- ☐ Not familiar
- ☐ Basic
- ☐ Intermediate
- ☐ Advanced

4. Have you worked with biomedical audio before? *

- ☐ Yes
- ☐ No
- ☐ Not sure

5. Which type of audio-related task are you most familiar with? *

- ☐ Audio recording
- ☐ Audio editing
- ☐ Audio classification
- ☐ Audio separation
- ☐ Machine learning model testing
- ☐ I have not worked with audio tasks before

Section B - Current Challenges in Cardiopulmonary AudioProcessing:

Help us understand the main problems in separating heart and lung sounds from mixed audio.

1. How difficult do you think it is to separate heart and lung sounds from a mixed cardiopulmonary recording? *

- ☐ Very easy
- ☐ Easy
- ☐ Neutral
- ☐ Difficult
- ☐ Very difficult

2. What problems do you think affect cardiopulmonary sound processing the most? *

- ☐ Background noise
- ☐ Overlap between heart and lung sounds
- ☐ Low recording quality
- ☐ Difficulty identifying useful sound components
- ☐ Lack of simple software tools
- ☐ Lack of clear output visualization
- ☐ Other:

3. Do you think noisy audio recordings can reduce the quality of machine learning-based sound processing? *

- ☐ Strongly disagree
- ☐ Disagree
- ☐ Neutral
- ☐ Agree
- ☐ Strongly Agree

4. Do you think a simple software prototype for heart-lung sound separation would be useful for students or researchers? *

- ☐ Strongly disagree
- ☐ Disagree
- ☐ Neutral
- ☐ Agree
- ☐ Strongly Agree

Section C - System Feature Preferences:

Choose the features you think would make this sound separation system more useful.

1. Which feature would be most important in the proposed system? *

- ☐ Upload mixed cardiopulmonary audio
- ☐ Preprocess noisy audio
- ☐ Separate heart and lung sounds
- ☐ Preview separated audio
- ☐ Download separated outputs
- ☐ View processing history
- ☐ View evaluation results

2. Which output would you prefer after sound separation? *

- ☐ Heart and lung audio files only
- ☐ Audio files with waveform visualization
- ☐ Audio files with spectrogram visualization
- ☐ Audio files with evaluation metrics
- ☐ All of the above

3. How important is it for the system to provide a preview of the separated heart and lung sounds? *

- ☐ Not important
- ☐ Slightly important
- ☐ Moderately important
- ☐ Important
- ☐ Very important

4. How important is it for the system to allow users to download separated audio outputs? *

- ☐ Not important
- ☐ Slightly important
- ☐ Moderately important
- ☐ Important
- ☐ Very important

5. How important is it for the system to show basic processing information, such as file name, processing status, and completion time? *

- ☐ Not important
- ☐ Slightly important
- ☐ Moderately important
- ☐ Important
- ☐ Very important

6. Should the system include a history page showing previously uploaded files and separation results? *

- ☐ Yes
- ☐ No
- ☐ Not sure

Section D - Interface and Output Preferences:

Tell us how you want the system to show results, audio previews, charts, and messages.

1. How would you prefer the separated results to be displayed? *

- ☐ Simple audio player only
- ☐ Audio player with waveform
- ☐ Audio player with spectrogram
- ☐ Audio player with waveform, spectrogram, and metrics
- ☐ Not sure

2. Which interface style would you prefer? *

- ☐ Simple interface with only main features
- ☐ Detailed interface with many technical options
- ☐ Balanced interface with simple view and optional advanced details

3. How important is it for the system interface to be easy for non-expert users? *

- ☐ Not important
- ☐ Slightly important
- ☐ Moderately important
- ☐ Important
- ☐ Very important

4. What information should appear first after separation is completed? *

- ☐ Download buttons for heart and lung sounds
- ☐ Audio preview players
- ☐ Waveform or spectrogram visualization
- ☐ Evaluation metrics
- ☐ Processing status summary

5. What type of error message would be most helpful if the uploaded file is invalid? *

- ☐ Short message only
- ☐ Message explaining the problem
- ☐ Message explaining the problem and suggested fix
- ☐ Not sure

Section E - Evaluation and Improvement:

Share what would make the system more accurate, useful, and easier to improve.

1. How important is it for the system to show evaluation metrics for the separated output? *

- ☐ Not important
- ☐ Slightly important
- ☐ Moderately important
- ☐ Important
- ☐ Very important

2. Which evaluation information would be most useful? *

- ☐ Signal quality score
- ☐ Separation accuracy score
- ☐ Noise reduction information
- ☐ Processing time
- ☐ All of the above

3. Would you use a prototype that separates mixed cardiopulmonary audio into heart and lung sound outputs? *

- ☐ Yes
- ☐ No
- ☐ Maybe

4. What improvement or feature would you like to see in a cardiopulmonary sound separation system?

Long answer text
